```
global_data = {
    "attribute_1": "total_cost",
    "attribute_2": "material_continuity",
    "attribute_3": "thermal_performance",
    "attribute_4": "mass",
    "attribute_5": "corrosion_resistance",

    "direction_attribute_1": "negative",   # lower cost        → better
    "direction_attribute_2": "positive",   # higher continuity → better
    "direction_attribute_3": "positive",   # higher efficiency → better
    "direction_attribute_4": "negative",   # lower mass        → better
    "direction_attribute_5": "positive",   # higher corr. res. → better

    "attribute_1_min": 50,    "attribute_1_max": 500,
    "attribute_2_min": 0,     "attribute_2_max": 1,
    "attribute_3_min": 0.50,  "attribute_3_max": 0.95,
    "attribute_4_min": 2.0,   "attribute_4_max": 25.0,
    "attribute_5_min": 1.0,   "attribute_5_max": 10.0,

    "copper_price_per_ton":   9500,   # baseline $/ton — varied in sweep
    "aluminum_price_per_ton": 2500,   # fixed $/ton throughout sweep
  "copper_price_per_kg": 9.500,     # fixed global Cu price
    "aluminum_price_per_kg": 2.500,   # fixed global Al price
}

regional_data = [
    {
        "region": "United States",
        "copper_product_market_share":   0.70,
        "aluminum_product_market_share": 0.30,
        "weight_attribute_1": 0.25,
        "weight_attribute_2": 0.20,
        "weight_attribute_3": 0.30,
        "weight_attribute_4": 0.10,
        "weight_attribute_5": 0.15,
    },
    {
        "region": "China",
        "copper_product_market_share":   0.40,
        "aluminum_product_market_share": 0.60,
        "weight_attribute_1": 0.40,
        "weight_attribute_2": 0.15,
        "weight_attribute_3": 0.20,
```

```
        "weight_attribute_4": 0.15,
        "weight_attribute_5": 0.10,
    },
    {
        "region": "India",
        "copper_product_market_share":   0.65,
        "aluminum_product_market_share": 0.35,
        "weight_attribute_1": 0.40,
        "weight_attribute_2": 0.10,
        "weight_attribute_3": 0.20,
        "weight_attribute_4": 0.10,
        "weight_attribute_5": 0.20,
    },
    {
        "region": "Europe",
        "copper_product_market_share":   0.55,
        "aluminum_product_market_share": 0.45,
        "weight_attribute_1": 0.20,
        "weight_attribute_2": 0.15,
        "weight_attribute_3": 0.35,
        "weight_attribute_4": 0.15,
        "weight_attribute_5": 0.15,
    },
]

# ── attribute_2_value assignment rule (applied below in product_data) ──────────
# For each region, identify which material is dominant (highest market share).
# The product matching that material → attribute_2_value = 1 (historical continuity)
# The product not matching          → attribute_2_value = 0 (break from history)
#
#   United States : Cu dominant (0.70) → Cu=1, Al=0
#   China         : Al dominant (0.60) → Cu=0, Al=1  ← KEY FIX
#   India         : Cu dominant (0.65) → Cu=1, Al=0
#   Europe        : Cu dominant (0.55) → Cu=1, Al=0

product_data = [
    {
        "region": "United States", "dominant material": "cu",
        "product": "Residential Copper Tube-and-Fin HVAC Coil",
        "copper_kg": 7.0, "aluminum_kg": 2.0,
        "non_material_cost_per_unit": 160,
        "attribute_2_value": 1,    # Cu matches US regional material history
        "attribute_3_value": 0.85,
        "attribute_4_value": 9.0,
```

```
    "attribute_5_value": 8.5,
},
{
    "region": "United States", "dominant material": "al",
    "product": "Residential Aluminum Microchannel HVAC Coil",
    "copper_kg": 0.5, "aluminum_kg": 6.0,
    "non_material_cost_per_unit": 160,
    "attribute_2_value": 0,    # Al breaks from US regional material history (Cu)
    "attribute_3_value": 0.80,
    "attribute_4_value": 6.5,
    "attribute_5_value": 6.0,
},

{
    "region": "China", "dominant material": "cu",
    "product": "Residential Copper Tube-and-Fin HVAC Coil",
    "copper_kg": 7.0, "aluminum_kg": 2.0,
    "non_material_cost_per_unit": 75,
    "attribute_2_value": 0,    # ← CORRECTED (was 1): Cu breaks from China's Al history
    "attribute_3_value": 0.85,
    "attribute_4_value": 9.0,
    "attribute_5_value": 8.5,
},
{
    "region": "China", "dominant material": "al",
    "product": "Residential Aluminum Microchannel HVAC Coil",
    "copper_kg": 0.5, "aluminum_kg": 6.0,
    "non_material_cost_per_unit": 75,
    "attribute_2_value": 1,    # ← CORRECTED (was 0): Al matches China's Al history
    "attribute_3_value": 0.80,
    "attribute_4_value": 6.5,
    "attribute_5_value": 6.0,
},

{
    "region": "India", "dominant material": "cu",
    "product": "Residential Copper Tube-and-Fin HVAC Coil",
    "copper_kg": 7.0, "aluminum_kg": 2.0,
    "non_material_cost_per_unit": 70,
    "attribute_2_value": 1,    # Cu matches India's regional material history
    "attribute_3_value": 0.85,
    "attribute_4_value": 9.0,
    "attribute_5_value": 8.5,
},
```

{
"region": "India", "dominant material": "al",
"product": "Residential Aluminum Microchannel HVAC Coil",
"copper_kg": 0.5, "aluminum_kg": 6.0,
"non_material_cost_per_unit": 70,
"attribute_2_value": 0,   # Al breaks from India's regional material history (Cu)
"attribute_3_value": 0.80,
"attribute_4_value": 6.5,
"attribute_5_value": 6.0,
},

{
"region": "Europe", "dominant material": "cu",
"product": "Residential Copper Tube-and-Fin HVAC Coil",
"copper_kg": 7.0, "aluminum_kg": 2.0,
"non_material_cost_per_unit": 150,
"attribute_2_value": 1,   # Cu matches Europe's regional material history
"attribute_3_value": 0.85,
"attribute_4_value": 9.0,
"attribute_5_value": 8.5,
},
{
"region": "Europe", "dominant material": "al",
"product": "Residential Aluminum Microchannel HVAC Coil",
"copper_kg": 0.5, "aluminum_kg": 6.0,
"non_material_cost_per_unit": 150,
"attribute_2_value": 0,   # Al breaks from Europe's regional material history (Cu)
"attribute_3_value": 0.80,
"attribute_4_value": 6.5,
"attribute_5_value": 6.0,
},
]